Supplementary material for "Exact characterization of three-terminal fermion counting under one-body scattering"

Zhanning Wang,[1,2] Yongtao Li,[1,2] Rubén Seoane Souto,[1,2] and José C. Abadillo-Uriel[1,2]

[1] *Quantum Advanced Research Center (QuARC) - Consejo Superior de Investigaciones Científicas (CSIC)*
[2] *Instituto de Ciencia de Materiales de Madrid (ICMM) - Consejo Superior de Investigaciones Científicas (CSIC) - Sor Juana Inés de la Cruz 3, 28049 Madrid, Spain*

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We first introduce how to calculate the counting probabilities for fermions passing through a fixed scatterer without interactions. We use this description to study when one scatterer can reproduce all the counting probabilities at three output terminals. For two particles, we express the result as four inequalities and apply them to toy models and recent experimental measurements. We show that either checking each terminal separately, or looking only at averages and pairwise correlations, are misleading. The results may seems to be consistent with one-body scattering, even when no single fixed scatterer can actually produce them. Finally, a four-terminal example shows why the three-terminal result does not extend to more terminals.

S1. TERMINAL EFFECTS AND THE COUNTING DETERMINANT

To determine whether a fixed scatterer can reproduce a measured counting distribution, we first derive how the counting probabilities depend on the occupied single particle orbitals after scattering. We consider $N \geq 1$ fermions sent to the scatter, and there are m output terminals. In each test, the fermions initially occupy N orthonormal single-particle orbitals $\{|\phi_i\rangle\}_{i=1}^{N}$, satisfying $\langle\phi_i|\phi_j\rangle = \delta_{ij}$. With one fermion occupying each orbital, the initial many-body state is a pure Slater determinant.

A fixed one-particle unitary scatterer operator S transforms these input orbitals into the output orbitals as $|\psi_i\rangle = S\,|\phi_i\rangle$. We collect the input and output orbitals into matrices $V = [|\phi_1\rangle, \ldots, |\phi_N\rangle]$ and $W = [|\psi_1\rangle, \ldots, |\psi_N\rangle] = SV$, respectively for use later. The one-particle Hilbert space contains all input and output modes.

The output modes are divided into m disjoint terminals, each consisting of a specified group of orthogonal modes [1]. For terminal i, we introduce the projection operator Q_i, so that $Q_i\,|\psi_r\rangle$ gives the component of output orbital $|\psi_r\rangle$ within that terminal. Because the terminals are disjoint, and account for every output mode, their projection operator and the associated $N \times N$ overlap matrices satisfy:

$$Q_iQ_j = \delta_{ij}Q_i\,, \quad \sum_{i=1}^{m} Q_i = \mathbb{1}\,, \quad A_i = W^\dagger Q_i W \succeq 0\,, \quad \sum_{i=1}^{m} A_i = \mathbb{1}_N\,. \tag{S1}$$

For a Hermitian matrix, $A \succeq 0$ means that all eigenvalues are nonnegative (positive semidefinite), and $A \succ 0$ means that they are all positive (positive definite). The δ_{ij} is the Kronecker symbol.

Since Q_i is an orthogonal projector, the matrix $A_i = W^\dagger Q_i W = (Q_iW)^\dagger(Q_iW)$ is positive semidefinite. Its matrix elements $(A_i)_{rs} = \langle\psi_r|Q_i|\psi_s\rangle$ describe the overlap between the occupied output orbitals $|\psi_r\rangle$ and $|\psi_s\rangle$ within terminal i. The output orbitals are surely orthogonal over the full space, but from here, we see that their components within a single terminal need not be orthogonal. For $i \neq j$, these local overlaps satisfy:

$$\sum_i \langle\psi_i|Q_i|\psi_j\rangle = \langle\psi_i|\psi_j\rangle = 0\,. \tag{S2}$$

The matrices A_i is called terminal effects. It is the local orbital overlaps that give rise to exchange interference in the terminal counting probabilities. Therefore, we demonstrate that for one-body operator S, we can get this terminal effect A. This transform the study of a huge matrix S into several much smaller size matrix A.

We now show the converse: if we have a set of matrices $A_i \succeq 0$ satisfying $\sum_i A_i = \mathbb{1}_N$, then this matrices can be realized by a fixed one-particle scatterer, provided that each terminal has sufficiently many output modes. To construct such a scatterer, we first build the occupied output orbitals $W = (|\psi_1\rangle, \ldots, |\psi_N\rangle)$ from the given matrices A_i. Consider mN output modes, with N modes assigned to each terminal, and let Q_i project onto the modes of terminal i. Since each A_i is positive semidefinite, it has a positive semidefinite square root $A_i^{1/2}$. We construct W by stacking these square roots:

$$W = \begin{bmatrix} A_1^{1/2} \\ A_2^{1/2} \\ \vdots \\ A_m^{1/2} \end{bmatrix} . \tag{S3}$$

This construction satisfies:

$$W^\dagger W = \sum_{i=1}^{m} A_i = \mathbb{1}_N, \quad W^\dagger Q_i W = A_i \, . \tag{S4}$$

Thus, the columns of W form N orthonormal output orbitals with exactly the prescribed terminal effects. Given any set of orthonormal input orbitals V in the same one-particle Hilbert space, we can extend the columns of V and W to complete orthonormal bases. A unitary transformation S mapping one basis to the other then satisfies $SV = W$, providing the required fixed scatterer.

The construction above uses mN output modes, but fewer modes may suffice. If terminal i contains only d_i modes, its effect must satisfy $\operatorname{rank} A_i \leq d_i$. This condition is also sufficient - we can factor $A_i = C_i^\dagger C_i$, where C_i is a $d_i \times N$ matrix, and construct W by stacking the matrices C_i. Stacking the C_i gives $W^\dagger W = I_N$ and $W^\dagger Q_i W = A_i$. Extending its orthonormal columns to a complete basis will yield a correct scatterer with d_i modes in terminal i.

So far, we know how to construct a fixed scatterer from a set of terminal effects A_i. We now need to relate the terminal effects A_i to the counting probabilities, which is the direct experimental data. Recall that each experiment starts from the same Slater determinant and undergoes the same one-particle scattering transformation S.

Let n_i be the occupation of terminal i and $P(\boldsymbol{n})$ the probability of the count vector $\boldsymbol{n} = (n_1, \ldots, n_m)$. The fixed particle number gives $\sum_i n_i = N$. The generating polynomial collects the probabilities using complex counting variables $\boldsymbol{z} = (z_1, \ldots, z_m)$:

$$\chi_N(\boldsymbol{z}) = \sum_{n_1+n_2+\cdots=N} P(\boldsymbol{n}) \prod_{i=1}^{m} z_i^{n_i} \, . \tag{S5}$$

Then the mean occupations is $\mu_i = \sum_{\boldsymbol{n}} n_i P(\boldsymbol{n}) = \langle n_i \rangle$. Here and below, expectation values refer to the counting distribution.

We now derive how the terminal effects A_i determine the counting polynomial $\chi_N(\boldsymbol{z})$. We assign a factor z_i to each particle occupying terminal i, so that a configuration with occupations $\boldsymbol{n}$ carries the weight $\prod_i z_i^{n_i}$. Let $\hat{c}^\dagger(\psi)$ create a fermion in orbital $|\psi\rangle$, and let $|0\rangle$ denote the vacuum. The output Slater state is $|\Psi_{\text{out}}\rangle = \hat{c}^\dagger(\psi_1) \cdots \hat{c}^\dagger(\psi_N) \, |0\rangle$.

We knew that any output state $|\psi_r\rangle$ can be decomposed using Q_i as $|\psi_r\rangle = \sum Q_i \, |\psi_r\rangle$, from where we define an auxiliary counting state $|\psi_r(\boldsymbol{z})\rangle = \sum z_i Q_i \, |\psi_r\rangle$. Then an unnormalized state is $|\Psi(\boldsymbol{z})\rangle = \hat{c}^\dagger(\psi_1(\boldsymbol{z})) \cdots \hat{c}^\dagger(\psi_N(\boldsymbol{z})) \, |0\rangle$. If we calculate the overlap with the original output state is exactly χ_N [2–6]:

$$\chi_N(\boldsymbol{z}) = \langle \Psi_{\text{out}} | \Psi(\boldsymbol{z}) \rangle = \det[\, \langle \psi_r | Z(\boldsymbol{z}) | \psi_s \rangle]_{r,s=1}^{N} = \det \left(\sum_{i=1}^{m} z_i A_i \right) . \tag{S6}$$

The coefficients are nonnegative probabilities, and normalization gives $\chi_N(\mathbf{1}) = 1$. Every term has total degree N, so the polynomial is homogeneous of degree N.

So far, the determinant representation provides a way to test whether a measured counting distribution is compatible with one-body scattering, without explicitly reconstructing the scatterer or the terminal effects. We now are going to show that every polynomial generated by a fixed one-body scatterer must satisfy a mathematical property known as real stability.

A polynomial $f(\boldsymbol{z})$ with real coefficients is real stable if $f(\boldsymbol{z}) \neq 0$ whenever $\operatorname{Im} z_i > 0$ for all i. To prove this property for one-body counting [7, 8], let $M(\boldsymbol{z}) = \sum_i z_i A_i$ with $z_i = x_i + \mathrm{i} y_i$, where x_i and y_i are real and all $y_i > 0$. Its Hermitian imaginary part satisfies:

$$\operatorname{Im} M = \frac{M - M^\dagger}{2\mathrm{i}} = \sum_i y_i A_i \succeq \left(\min_i y_i \right) \mathbb{1}_N \succ 0 \, . \tag{S7}$$

If $M\boldsymbol{v} = 0$ for some nonzero $\boldsymbol{v}$, taking the imaginary part of $\boldsymbol{v}^\dagger M \boldsymbol{v} = 0$ contradicts this strict positivity. It follows that $\chi_N(\boldsymbol{z}) \neq 0$ whenever all counting variables have positive imaginary parts. Then we conclude that fixed one-body scattering nature of S must require $\chi_N(\boldsymbol{z})$ is real stable.

S2. COMPLETE THREE-TERMINAL REALIZABILITY

So far, we have established that any fixed one-body scatterer produces a real-stable counting polynomial. This only gives a necessary condition for one-body scattering. The next question is that if a measured counting distribution can be mapped to real stable polynomial, can we always find a single scatterer that reproduces all its probabilities?

For three output terminals, the answer is yes! To prove this, we start from an arbitrary three-terminal counting distribution for a fixed number of particles $N \geq 1$, and denote its generating polynomial by $f(z_1, z_2, z_3)$. Since its coefficients are probabilities and the total particle number is fixed, f is homogeneous of degree N, has nonnegative coefficients, and satisfies $f(1,1,1) = 1$. We now assume that f is real stable and aim to show that there exist positive semidefinite matrices A_1, A_2, A_3, summing to the identity, whose determinant reproduces f.

Our point is that, once these terminal effects are found, the construction in Section S1 gives the required fixed scatterer. To obtain a suitable determinant representation of f, we must consult the Helton–Vinnikov theorem [9].

We first examine the roots of a polynomial when its arguments vary along a real line. A real homogeneous polynomial is hyperbolic with respect to a real vector $\boldsymbol{e}$ if $f(\boldsymbol{e}) > 0$ and $t \mapsto f(\boldsymbol{x}+t\boldsymbol{e})$ has only real roots for every real $\boldsymbol{x}$ [10]. Take $\boldsymbol{e} = (1,1,1)^T$. Stability excludes roots with $\operatorname{Im} t > 0$, and real coefficients exclude roots with $\operatorname{Im} t < 0$ by complex conjugation. Because f is homogeneous and normalized, the polynomial in t has degree N and leading coefficient one. The polynomial f is hyperbolic with respect to $\boldsymbol{e}$.

HV theorem then supplies real symmetric $N \times N$ matrices whose sum is positive definite [9, 11]. After absorbing a positive scalar into the matrices, the determinant of their linear combination reproduces the polynomial:

$$f(\boldsymbol{z}) = \det\left(\sum_{i=1}^{3} z_i B_i\right), \quad B_i = B_i^T, \quad H := \sum_{i=1}^{3} B_i \succ 0\,. \tag{S8}$$

Repeated factors are allowed, as are singular zeros where all first derivatives vanish.

To use these matrices as terminal effects, their sum must be the identity and each must be positive semidefinite. The positive square root of H gives the required normalization:

$$A_i = H^{-1/2} B_i H^{-1/2}, \quad \sum_i A_i = \mathbb{1}_N, \quad \det H = f(\mathbf{1}) = 1, \quad f(\boldsymbol{z}) = \det\left(\sum_i z_i A_i\right). \tag{S9}$$

We now need show that each A_i is positive semidefinite.

Let $\boldsymbol{e}_i$ be the coordinate vector with entry one at position i and zero elsewhere. Increasing one counting variable from one, while leaving the others equal to one, cannot decrease the polynomial because its coefficients are nonnegative. For every real $t \geq 0$, this gives:

$$\det(\mathbb{1}_N + tA_i) = f(\mathbf{1} + t\boldsymbol{e}_i) \geq f(\mathbf{1}) = 1\,. \tag{S10}$$

A negative eigenvalue λ of the real symmetric matrix A_i would make this determinant vanish at $t = -1/\lambda > 0$. Every terminal effect therefore satisfies $A_i \succeq 0$. Then we finish constructing the required orthonormal output orbitals and unitary scatterer. In summary, we establish the converse of the result in Sec. S1.

For three terminals, real stability is both necessary and sufficient for a normalized counting distribution to admit a fixed one-body realization. The proof shows that any such real stable counting polynomial can be represented by positive semidefinite terminal effects that sum to the identity, and hence reproduced by a single fixed scatterer. This equivalence holds for arbitrary fixed particle number N.

While this gives a complete mathematical characterization, checking real stability directly may be inconvenient for experimentally measured distributions. This encourage us to turn to the two-particle case, where the realizability condition can be expressed directly in terms of the counting probabilities and given a simple geometric interpretations.

S3. THE COMPLETE TWO-PARTICLE CRITERION AND ITS GEOMETRY

For two fermions, we can decide whether a fixed scatterer reproduces a histogram directly from its probabilities. We first derive a matrix criterion valid for any number m of terminals, then express it as four scalar inequalities when $m = 3$.

Let $p_i = P(n_i = 2)$ be the probability that both particles reach terminal i, and let $s_{ij} = s_{ji} = P(n_i = n_j = 1)$ for distinct terminals i, j. All these probabilities are nonnegative, and satisfy $\sum_i p_i + \sum_{i<j} s_{ij} = 1$. In our theory, they

are the only variable required, which can be read directly from experimental results. Recall our goal is to determine whether these measured probabilities admit the determinant representation. Writing the generating polynomial as a quadratic form allows us to organize the probabilities into a matrix and test their compatibility with one-body scattering.

Their generating polynomial defines a real symmetric coefficient matrix G:

$$\chi_2(\boldsymbol{z}) = \sum_i p_i z_i^2 + \sum_{i<j} s_{ij} z_i z_j = \boldsymbol{z}^T G \boldsymbol{z}, \quad G_{ii} = p_i, \quad G_{ij} = \frac{s_{ij}}{2} \quad (i \neq j)\,. \tag{S11}$$

To find terminal effects that reproduce G, we write each candidate 2×2 Hermitian matrix using the identity and the Pauli matrices. For two particles, this decomposition is particularly useful because it converts the positivity and normalization conditions on the unknown terminal effects into simple constraints on real scalars and three-dimensional vectors.

Here, $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ denotes the three Pauli matrices, use $a_i = \operatorname{tr} A_i/2$ to multiply the identity, and use $\boldsymbol{r}_i \in \mathbb{R}^3$ as the three Pauli coefficients. The effects are positive semidefinite and sum to the identity exactly when:

$$A_i = a_i \mathbb{1}_2 + \boldsymbol{r}_i \cdot \boldsymbol{\sigma}, \quad a_i \geq |\boldsymbol{r}_i|, \quad \sum_i a_i = 1, \quad \sum_i \boldsymbol{r}_i = 0\,. \tag{S12}$$

The vector $\boldsymbol{r}_i$ is the Bloch vector of the effect in the two-dimensional occupied orbital space. The eigenvalues $a_i \pm |\boldsymbol{r}_i|$ give the arrival probabilities at terminal i for the two orbitals in the eigenbasis of A_i. Collect the scalar coefficients in $\boldsymbol{a} = (a_1, \ldots, a_m)^T$. Matching coefficients with the determinant identity $\det(a\mathbb{1}_2 + \boldsymbol{r} \cdot \boldsymbol{\sigma}) = a^2 - \boldsymbol{r} \cdot \boldsymbol{r}$ gives:

$$p_i = a_i^2 - |\boldsymbol{r}_i|^2, \quad s_{ij} = 2(a_i a_j - \boldsymbol{r}_i \cdot \boldsymbol{r}_j), \quad \boldsymbol{a} = G\mathbf{1}, \quad a_i = p_i + \frac{1}{2}\sum_{j \neq i} s_{ij} = \frac{\mu_i}{2}\,. \tag{S13}$$

The measured probabilities fix the inner products required of the Bloch vectors. Importantly, the relation $a_i = \mu_i/2$ shows that the scalar part of each effect is already fixed by the measured histogram.

Then we finished the first part of the proof, we know that from probability histogram, we can find $|\boldsymbol{r}_i$, and $\boldsymbol{r}_i \cdot \boldsymbol{r}_j$, i.e., the magnitude and angle. The remaining question is whether the Bloch vectors can be constructed, which call the Gram matrix out. The candidate Gram matrix required by our inner products can be constructed as:

$$R = \boldsymbol{a}\boldsymbol{a}^T - G, \quad R_{ij} = \boldsymbol{r}_i \cdot \boldsymbol{r}_j, \quad R\mathbf{1} = 0\,. \tag{S14}$$

Now we can to allow for the existance of $\boldsymbol{r}_i$, what condition should R statisfy.

Assuming $\boldsymbol{r}_i$ can exist, then we can calculate, for any real vector $\boldsymbol{x}$:

$$\boldsymbol{x}^T R \boldsymbol{x} = \left| \sum_i x_i r_i \right|^2 \geq 0\,, \tag{S15}$$

which means $R \succeq 0$, proving the necessity. We need to prove that $R \succeq 0$ can support the construction of the scatter.

From linear algebra, we know that if a real symmetric matrix R is positive semidefinite, and rand $R \leq 3$, then there must exist $\boldsymbol{r}_i$ satisfying $R_{ij} = \boldsymbol{r}_i \cdot \boldsymbol{r}_j$, with $R\mathbf{1} = 0$. This proves that as long as $R \succeq 0$, $\boldsymbol{r}_i$ can be constructed. Now we can further check:

$$\|\boldsymbol{r}_1 + \boldsymbol{r}_2 + \boldsymbol{r}_3\|^2 = \sum_{i,j} \boldsymbol{r}_i \cdot \boldsymbol{r}_j = 0 \tag{S16}$$

So we do have $\boldsymbol{r}_1 + \boldsymbol{r}_2 + \boldsymbol{r}_3 = 0$. This gives the complete criterion:

$$\text{one-body realizability} \quad \Leftrightarrow \quad R \succeq 0 \text{ and } \quad \operatorname{rank} R \leq 3\,. \tag{S17}$$

This reformulation turns the search for an unknown scatterer into a positivity and rank test on a matrix determined entirely by the measured counting probabilities.

However, we notice that so far we have not connect to the proability histrogram directly yet. To obtain inequalities directly in the six probabilities, we replace the Gram matrix test by a condition on the eigenvalues of G. In the following, we would like to prove that for a normalized two-particle distribution, $R \succeq 0$ is equivalent to G having exactly one positive eigenvalue, with any remaining eigenvalues nonpositive [12].

First, we consider all the real vectors perpendicular to $\boldsymbol{a}$, i.e., $\boldsymbol{a}\cdot\boldsymbol{x}=0$. From $R=\boldsymbol{a}\boldsymbol{a}^T-G$, we have $\boldsymbol{x}^TG\boldsymbol{x}=(\boldsymbol{a}\cdot\boldsymbol{x})^2-\boldsymbol{x}^TR\boldsymbol{x}$ giving $\boldsymbol{x}^TG\boldsymbol{x}=-\boldsymbol{x}^TR\boldsymbol{x}\le 0$. Therefore, in the subsapce orthogonal to $\boldsymbol{a}$, the quadratic form of G is nonnegative, meaning there can exist at most one postive eigenvalue. Furthermore, from normalization $\mathbf{1}^TG\mathbf{1}=1>0$, we deduce that there must be at least one positive eigenvalue.

Next, we assuming G as at most one positive eigenvalue, for any $\boldsymbol{y}$ in the subsapce orthogonal to $\boldsymbol{a}$, we have $\boldsymbol{a}\cdot\boldsymbol{y}=0$ again, Since $G\mathbf{1}=\boldsymbol{a}$, then $\mathbf{1}^TGy=\boldsymbol{a}\cdot\boldsymbol{y}=0$. If $\boldsymbol{y}^TG\boldsymbol{y}>0$, then from $\mathbf{1}^TG\mathbf{1}1>0$ and $\mathbf{1}^TG\boldsymbol{y}=0$, we deduce that the quadratic form in the two-dimensional space spanned by $\mathbf{1}$ and $\boldsymbol{y}$ must be postive definite, contradicting the assumption of at most one positive eigenvalue, therefore $\boldsymbol{y}^TG\boldsymbol{y}\le 0$. Doing the standard decompsition to arbitary vector $\boldsymbol{x}$, we get $\boldsymbol{x}=(\boldsymbol{a}\cdot\boldsymbol{x})\mathbf{1}+\boldsymbol{y}$, then we have:

$$\boldsymbol{x}^TR\boldsymbol{x}=(\boldsymbol{a}\cdot\boldsymbol{x})^2-\boldsymbol{x}^TG\boldsymbol{x}=-\boldsymbol{y}^TG\boldsymbol{y}\ge 0\,. \tag{S18}$$

Therefore we porved $R\succeq 0$, and closed the proof chain as:

$$\text{one-body realizability}\Leftrightarrow R\succeq 0\Leftrightarrow G\text{ has exactly one postive eigenvalue} \tag{S19}$$

For $m=3$, this eigenvalue condition is equivalent to four polynomial inequalities:

$$\Delta_{ij}=s_{ij}^2-4p_ip_j\ge 0\quad(1\le i<j\le 3)\,,\quad \Delta_3=4p_1p_2p_3+s_{12}s_{13}s_{23}-p_1s_{23}^2-p_2s_{13}^2-p_3s_{12}^2\ge 0\,. \tag{S20}$$

We first derive the first three inequalities. Consider a principal minors G_{ij} of G:

$$G_{ij}=\begin{bmatrix}p_i & s_{ij}/2\\ s_{ij}/2 & p_j\end{bmatrix}. \tag{S21}$$

If it has two eigenvalues, then the quadratic form of G will be nonnegative, contradicting with single positive eigenvalue nature of G. Beside G_{ij} cannot have two positive eigenvalues, we can prove it cannot have two nagative eigenvalues as well, by the fact that $\operatorname{tr}G_{ij}=p_i+p_j>0$. Then we conclude $\det G_{ij}\le 0$ which translate to $p_ip_j-s_{ij}^2/4\le 0$. To see the fourth one, recall that G has three eigenvalues $\lambda_1>0$, $\lambda_2\le 0$, and $\lambda_3\le 0$, then we have $\det G=\lambda_1\lambda_2\lambda_3\ge 0$, which can be rephrased into the inequality.

Conversely, the four inequalities rule out two or three positive eigenvalues. To prove this, we can check the sum of their pairwise products is fixed by the principal minors:

$$\lambda_1\lambda_2+\lambda_1\lambda_3+\lambda_2\lambda_3=-\frac{\Delta_{12}+\Delta_{13}+\Delta_{23}}{4}\le 0\,. \tag{S22}$$

We can deduce that three positive eigenvalues contradict this inequality. Two positive eigenvalues with a negative third contradict $\det G\ge 0$, whereas two positive eigenvalues with a zero third again make the pairwise-product sum positive. Because normalization guarantees a positive eigenvalue, exactly one remains, proving sufficiency. Therefore, the four inequalities provide a complete criterion directly in terms of the measured probabilities.

For three terminals, the criterion has a simple geometric meaning that the three Bloch vectors must form a closed triangle. Their lengths are fixed by the measured probabilities, $\ell_i=\sqrt{a_i^2-p_i}$, and satisfy $\ell_i\le a_i$ because $p_i\ge 0$. Then the question becomes to determine if we can form a triangle with three length (from histogram). Now we show that, a fixed one-body scatterer exists if and only if these lengths are real and satisfy the triangle inequalities.

To see sufficiency, choose a triangle with those lengths and assign its oriented sides to $\boldsymbol{r}_i$. Since the three side vectors sum to zero, every off-diagonal inner product is $(\ell_k^2-\ell_i^2-\ell_j^2)/2$ for distinct i,j,k. This equals the required entry R_{ij} because $R\mathbf{1}=0$, so the constructed vectors recover all six probabilities. The necessity is trivial since $\sum\boldsymbol{r}_i=0$.

The triangle area also tells us whether all three effects can be diagonalized in the same occupied orbital basis. To determine this from the measured probabilities, we express the area in terms of Δ_3. Choose a plane containing the three vectors, write their coordinates as (r_{ix},r_{iy}), and let B have rows (a_i,r_{ix},r_{iy}). Then $G=B\,\mathrm{diag}(1,-1,-1)B^T$. Adding the three rows of B produces $(1,0,0)$, so $\det B$ equals the oriented parallelogram area of any pair of Bloch vectors, up to a sign. The determinant measures the squared area:

$$\det G=R_{ii}R_{jj}-R_{ij}^2=|\boldsymbol{r}_i\times\boldsymbol{r}_j|^2,\quad \Delta_3=4|\boldsymbol{r}_i\times\boldsymbol{r}_j|^2=16\mathcal{A}^2\quad(i\ne j)\,, \tag{S23}$$

where $\mathcal{A}$ is the triangle area.

Now we prove that the area of the triangle, dictates if the terminal effect A_i can be diagonalized simultaneously, and if two terminal effects are commuting, then the corresponding Bloch vectors will be collinear. Firstly, we notice that $[A_i,A_j]=2\mathrm{i}(\boldsymbol{r}_i\times\boldsymbol{r}_j)\cdot\boldsymbol{\sigma}$, meaning that $[A_i,A_j]=0$ is equivalent to $\boldsymbol{r}_i\times\boldsymbol{r}_j=0$. Next, we measure the commutator

using the Frobenius norm, defined for a matrix C by $\|C\|_{\mathrm{F}}^2 = \mathrm{tr}\big(C^\dagger C\big)$:

$$\|[A_i, A_j]\|_{\mathrm{F}}^2 = 8|\boldsymbol{r}_i \times \boldsymbol{r}_j|^2 = 2\Delta_3\,. \tag{S24}$$

For a histogram that satisfies the four inequalities, $\Delta_3 = 0$ makes all Bloch vectors lie on one line. The effects then commute and can be simultaneously diagonalized.

The significance of simultaneous diagonalization is that all terminal effects share a common orbital basis in which the off-diagonal overlaps vanish. In this basis, the matrix entering the counting determinant is diagonal, so its determinant factorizes into two linear generating polynomials. Each factor describes a normalized single-particle arrival distribution, and their product represents the counting statistics of two independent arrivals.

Let u_i and v_i be their diagonal entries in that common basis, so they give the probabilities for two independent single-particle arrivals at terminal i. The generating polynomial then splits into two single-particle factors:

$$\chi_2(\boldsymbol{z}) = \left(\sum_i u_i z_i\right)\left(\sum_i v_i z_i\right), \quad u_i, v_i \geq 0, \quad \sum_i u_i = \sum_i v_i = 1\,. \tag{S25}$$

Conversely, the factorization in Eq. (S25) supplies the diagonal effects $A_i = \mathrm{diag}(u_i, v_i)$ and gives $\Delta_3 = 0$. An allowed histogram with $\Delta_3 > 0$ requires noncommuting terminal effects in every realization. In summary, we have $\Delta_3 = 0$ if and only if the independent arrival description exists, and $\Delta_3 > 0$ indicates that noncommuting effects are necessary.

We next examine what information is lost when the counting distribution is tested only through partial terminal statistics. The separate two-terminal tests can arise from two different operations. First, one can retaining events in which a terminal is empty, or second, one can combine terminals together. The previous one removes some test results, while the later one keeps all the results but reduce categories.

Let us look at the first case. For distinct i, j, k among the three terminals, setting $z_k = 0$ removes all counting outcomes with $n_k > 0$, so we are only left with p_i, p_j, and s_{ij}. Their total probability is $h_{ij} = P(n_k = 0) = p_i + p_j + s_{ij}$. The normalized generating polynomial becomes to:

$$\chi_{ij|n_k=0}(z_i, z_j) = \frac{p_i z_i^2 + s_{ij} z_i z_j + p_j z_j^2}{h_{ij}}\,. \tag{S26}$$

To determine the one-body constraint on this distribution, recall that for two terminals the corresponding effects must sum to the identity and therefore commute [13, 14]. They can be simultaneously diagonalized, so a normalized two-particle one-body counting polynomial takes the form:

$$\chi_2(z_i, z_j) = (uz_i + (1-u)z_j)(vz_i + (1-v)z_j)\,, \tag{S27}$$

where $0 \leq u, v \leq 1$. Setting $z_j = 1$, the discriminant of the resulting quadratic polynomial is $(u-v)^2 \geq 0$, indicating that as long as a two-terminal distribution is from one-body scattering, we must get a nonnegative discriminnat. Conversely, any normalized quadratic counting polynomial with nonnegative coefficients and nonnegative discriminant admits such a factorization. Thus, the discriminant provides a necessary and sufficient condition for two-terminal one-body realizability.

Applying this condition to our distribution gives:

$$D_{ij} = \frac{s_{ij}^2 - 4p_i p_j}{h_{ij}^2} = \frac{\Delta_{ij}}{h_{ij}^2}\,. \tag{S28}$$

Since $h_{ij} > 0$, the conditional distribution admits a two-terminal one-body realization exactly when $\Delta_{ij} \geq 0$. In summary, the three inequalities $\Delta_{ij} \geq 0$ test the one-body compatibility of the three conditional two-terminal distributions separately. However, we emphasize that satisfying all three conditions does not guarantee a common three-terminal realization - we also need $\Delta_3 \geq 0$.

We now consider the second way of reducing the three-terminal counting distribution. We keep all events but count only the particles arriving at terminal i. The remaining terminals are treated as a single terminal, containing $N - n_i$ particles. To obtain the counting distribution for n_i, we set $z_i = t$ and all other counting variables to one. For a fixed one-body scatterer, this gives:

$$\chi_N(1, \ldots, t, \ldots, 1) = \det[\mathbb{1}N + (t-1)A_i] = \prod r = 1^N(1 - \nu_{ir} + \nu_{ir}t)\,, \tag{S29}$$

where ν_{ir} are the eigenvalues of A_i. Since $0 \preceq A_i \preceq \mathbb{1}N$, these eigenvalues satisfy $0 \leq \nu ir \leq 1$. Each factor describes a Bernoulli trial, where an arrival with probability ν_{ir}, or no arrival otherwise [13–15].

Returning to $N = 2$, define $b_i = P(n_i = 1) = \sum_{j\neq i} s_{ij}$ and $c_i = P(n_i = 0) = 1 - p_i - b_i$. The generating polynomial $p_i t^2 + b_i t + c_i$ for n_i has discriminant:

$$D_i = b_i^2 - 4p_i c_i = 4(a_i^2 - p_i) = 4R_{ii}\,. \tag{S30}$$

Each single-terminal discriminant only checks whether one squared length $a_i^2 - p_i$ is nonnegative. The three-terminal criterion also requires these lengths to form a closed triangle.

One-body scattering also limits pair correlations in a way that counts in distinct terminals cannot be positively correlated [2, 4, 7]. Define $\mathrm{Cov}(n_i, n_j) = \langle n_i n_j\rangle - \mu_i\mu_j$. For distinct terminals, differentiating the logarithm of Eq. (S6) at $\boldsymbol{z} = \mathbf{1}$ gives:

$$\mathrm{Cov}(n_i, n_j) = -\,\mathrm{tr}(A_i A_j) \leq 0 \quad (i \neq j)\,. \tag{S31}$$

Since A_i, and A_j are positive semidefinite matrices, we finished the proof. For two particles, it becomes $s_{ij} - 4a_i a_j \leq 0$, giving a pair-correlation test.

These single-terminal and pair-correlation conditions provide useful necessary tests, but they do not guarantee that the full counting distribution can be reproduced by a common set of terminal effects. The joint three-terminal inequalities provide the complete criterion. In the following sections, we apply this criterion to explicit scattering models and experimental counting distributions, examining both the existence of a one-body realization and the role of exchange interference.

S4. EXAMPLES OF THREE-TERMINAL COUNTING

The three-terminal criterion allows us to distinguish between counting distributions that admit a fixed one-body realization, those that can be described by independent arrival channels, and those that cannot arise from any fixed one-body scatterer. We illustrate these distinctions with two examples. The first shows how exchange interference produces noncommuting terminal effects even without interactions. The second shows that acceptable single-terminal statistics and pair correlations do not guarantee a common one-body realization.

We apply the criterion to a symmetric scatterer to test whether exchange alone can prevent independent arrivals [8]. Take a three-mode scatterer [16] with its first two input modes occupied:

$$S_{jr} = \frac{1}{\sqrt{3}} \mathrm{e}^{2\pi \mathrm{i}(j-1)(r-1)/3}, \quad j, r = 1, 2, 3\,. \tag{S32}$$

Here j labels an output mode and r labels an input mode. Each terminal contains one output mode, so Pauli exclusion gives $p_i = 0$. The two-by-two determinants formed from the occupied columns give the two-particle amplitudes; their squared moduli yield:

$$(p_1, p_2, p_3; s_{12}, s_{13}, s_{23}) = (0, 0, 0; 1/3, 1/3, 1/3), \quad \Delta_3 = \frac{1}{27}\,. \tag{S33}$$

The positive Δ_3 requires noncommuting terminal matrices and rules out independent single-particle arrivals, even though the particles do not interact.

We next ask whether another scatterer could give a larger Δ_3. For any realizable distribution, the triangle perimeter obeys $L_\triangle = \ell_1 + \ell_2 + \ell_3 \leq \sum_i a_i = 1$. The area is maximal for an equilateral triangle at fixed perimeter, so Eq. (S23) implies:

$$\mathcal{A} \leq \frac{L_\triangle^2}{12\sqrt{3}}, \quad 0 \leq \Delta_3 = 16\mathcal{A}^2 \leq \frac{1}{27}\,. \tag{S34}$$

The Fourier example reaches this maximum with $a_i = \ell_i = 1/3$.

To test whether the mean occupations alone can determine which histograms are possible, we vary the pair probabilities while keeping $\mu_i = 2/3$ in every terminal. In Fig. S1(c), we set $p_1 = 0$, $x = p_2$, and $y = p_3$, with $0 \leq x, y \leq 1/9$. The three means $\mu_i = 2/3$ determine:

$$s_{12} = \frac{1}{3} - x + y, \quad s_{13} = \frac{1}{3} + x - y, \quad s_{23} = \frac{1}{3} - x - y\,. \tag{S35}$$

Every point in this square has nonnegative probabilities. Each terminal separately has an allowed counting distribution, and every cross covariance is nonpositive. The distribution of the count in terminal i alone has generating polynomial $g_i(t)$. To compare these separate checks with the full criterion, we calculate its discriminant D_i and the two joint constraints that can change sign in this square:

$$g_i(t) = p_i t^2 + \left(\frac{2}{3} - 2p_i\right) t + \frac{1}{3} + p_i, \quad D_i = 4\left(\frac{1}{9} - p_i\right), \tag{S36}$$

$$\Delta_3(x,y) = \frac{1}{27} - \frac{2}{9}(x+y) - (x-y)^2, \tag{S37}$$

$$\Delta_{23}(x,y) = \frac{1}{9} - \frac{2}{3}(x+y) + (x-y)^2. \tag{S38}$$

The other two inequalities have squares on their left-hand sides and therefore always hold in this square. The Fourier point is $F = (0,0)$. At $X = (1/9, 1/36)$, all three tests restricted to events with one empty terminal pass, but $\Delta_3 < 0$ rules out a common scatterer. Section S7 explains why this happens. At $C = (1/9, 0)$, $\Delta_3 = 0$ and the terminal matrices commute. To verify that independent arrivals can produce C, choose the diagonal terminal matrices:

$$A_1 = \mathrm{diag}(2/3, 0), \quad A_2 = \mathbb{1}_2/3, \quad A_3 = \mathrm{diag}(0, 2/3). \tag{S39}$$

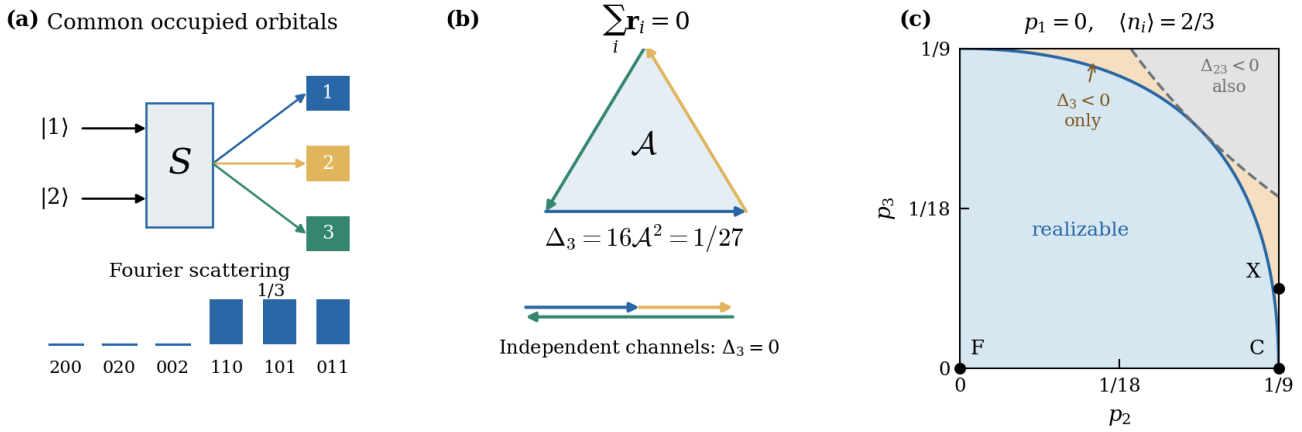


FIG. S1. Two-particle geometry and the condition for a common scatterer. (a) Two occupied input orbitals scatter into three terminals. The six bars show the balanced Fourier example, with $p_i = 0$ and $s_{ij} = 1/3$. (b) The Bloch vectors of the terminal matrices form a closed triangle of area $\mathcal{A}$, with $\Delta_3 = 16\mathcal{A}^2$. The equilateral Fourier triangle attains $\Delta_3 = 1/27$, while collinear vectors give commuting effects and independent arrival channels. (c) Distributions with $p_1 = 0$ and fixed means $\langle n_i \rangle = 2/3$. The solid curve is $\Delta_3 = 0$ and the dashed curve is $\Delta_{23} = 0$. A common scatterer exists in the blue region; in the beige region only $\Delta_3 \geq 0$ fails; in the gray region $\Delta_{23} \geq 0$ also fails. All single-terminal tests pass throughout the square. The labeled points are the Fourier example $F = (0,0)$, the commuting example $C = (1/9, 0)$, and the excluded histogram $X = (1/9, 1/36)$ from Eq. (S59).

S5. EXCHANGE INTERFERENCE AND NONCOMMUTING TERMINAL EFFECTS

The example above showed that a counting distribution can arise from fixed one-body scattering without admitting an independent-arrival description. We now ask whether the same distinction can be identified in experimental data. For this purpose, we analyze the two-particle momentum correlations reported in Ref. [17]. In this interference experiment, two fermions in the same internal state are released from separate tweezers and expand without interacting. Their momenta are resolved after a common time of flight. We estimate the counting histogram from the published two-particle correlation map in their Fig. 2(a).

Unlike the discrete output terminals considered so far, this experiment resolves a continuous momentum distribution. To apply our criterion, we divide the momentum modes into three disjoint groups and treat them as effective terminals. These groups define the orthogonal one-particle projectors. Let k denote the measured momentum and k_{lat} the momentum scale used on the original published axes. We introduce the dimensionless momentum $q = k/k_{\mathrm{lat}}$ and

allow the sector origin to shift by ϕ:

$$T_i(\phi) = \{q : 2(i-1)/3 \le (q-\phi) \bmod 2 < 2i/3\}, \quad i = 1, 2, 3\,. \tag{S40}$$

Each terminal keeps the momentum components in T_i and removes those outside it. This selection defines a one-particle projector. The three projectors are orthogonal and together cover the momentum axis; each sector contains many momentum modes.

Having defined the three effective terminals, we next need to obtain the six counting probabilities from the continuous momentum measurement. We do this by estimating the two-particle correlation function from the published map and integrating it over pairs of momentum sectors. The map shows the normal-ordered correlator [17]:

$$C(q_1, q_2) = \langle \hat{\psi}^\dagger(q_1)\hat{\psi}^\dagger(q_2)\hat{\psi}(q_2)\hat{\psi}(q_1)\rangle\ . \tag{S41}$$

Its integral over all momenta would equal $N(N-1) = 2$ for an ideal two-particle sample.

The displayed map covers a finite window, so we normalize the estimated correlator $\widetilde{C}$ over that window to obtain the pair density used for counting:

$$\Omega(L) = [-L, L]^2, \quad Z_L = \int_{\Omega(L)} \widetilde{C}(q_1, q_2)\, \mathrm{d}q_1\, \mathrm{d}q_2\,, \quad \rho_L(q_1, q_2) = \frac{\widetilde{C}(q_1, q_2)}{Z_L}\,. \tag{S42}$$

Here L is the inferred half-width of the displayed dimensionless momentum axis; we use $L = 3.7$. We estimate $\widetilde{C}$ using the intensity scale and exchange symmetry of the published map.

We obtain the six counting probabilities by integrating ρ_L over pairs of sectors. Let J_{ij} denote the ordered probability with the vertical momentum q_2 in sector i and the horizontal momentum q_1 in sector j:

$$J_{ij} = \int_{T_i \cap [-L,L]} \int_{T_j \cap [-L,L]} \rho_L(q_1, q_2)\, \mathrm{d}q_1\, \mathrm{d}q_2\,, \quad p_i = J_{ii}, \quad s_{ij} = J_{ij} + J_{ji} \quad (i < j)\,. \tag{S43}$$

Thus p_i is the probability of two particles in terminal i, and s_{ij} is the probability of one particle in each of terminals i and j. The two orientations of a split pair contribute to the same counting outcome. To test one-body realizability at $\phi = 0$ and $L = 3.7$, we evaluate the four quantities in Eq. (S20) using the histogram before rounding:

$$(p_1, p_2, p_3; s_{12}, s_{13}, s_{23}) \simeq (0.0506, 0.0535, 0.0472; 0.2880, 0.2839, 0.2768)\,, \tag{S44}$$

$$(\Delta_{12}, \Delta_{13}, \Delta_{23}; \Delta_3) \simeq (0.0721, 0.0710, 0.0665; 0.0110)\,. \tag{S45}$$

All four inequalities are satisfied, so the estimated histogram is compatible with a fixed one-body scatterer.

This establishes one-body realizability, but leaves open whether independent single-particle arrivals could reproduce the same counting probabilities. To distinguish these possibilities, we use the relation between Δ_3 and the commutators of the terminal effects:

$$\|[A_i, A_j]\|_{\mathrm{F}}^2 = 2\Delta_3 \simeq 0.0221 \quad (i \neq j)\,. \tag{S46}$$

Here $\|\cdot\|_F$ denotes the Frobenius norm. Since the commutator norm is nonzero, the terminal effects cannot be simultaneously diagonalized in any compatible one-body realization.

Because the terminal partition and momentum calibration enter the reconstruction of the counting probabilities, we check whether the inferred one-body compatibility and noncommutativity depend on these analysis choices. At $L = 3.7$, scanning the sector origin over a full period, $\phi \in [0, 2)$, gives $0.0109 \le \Delta_3 \le 0.0119$ and a minimum Δ_{ij} of 0.0639. We also vary the inferred axis half-width over $L \in [3.55, 3.85]$, recalibrating the momenta across the same full map for each value of L. The minimum values of Δ_3 and Δ_{ij} are then 0.0103 and 0.0621, respectively, so every sampled partition remains compatible and requires noncommuting effects. The scans in Fig. S2 test how the terminal definition and momentum calibration affect the conclusion.

Throughout the tested sector partitions and momentum calibrations, the estimated histograms remain compatible with fixed one-body scattering while requiring noncommuting terminal effects. This provides an experimental-data example of the distinction illustrated by the scatterer in Sec. S4. We next consider a different situation, in which the counting probabilities themselves exclude any fixed one-body realization.

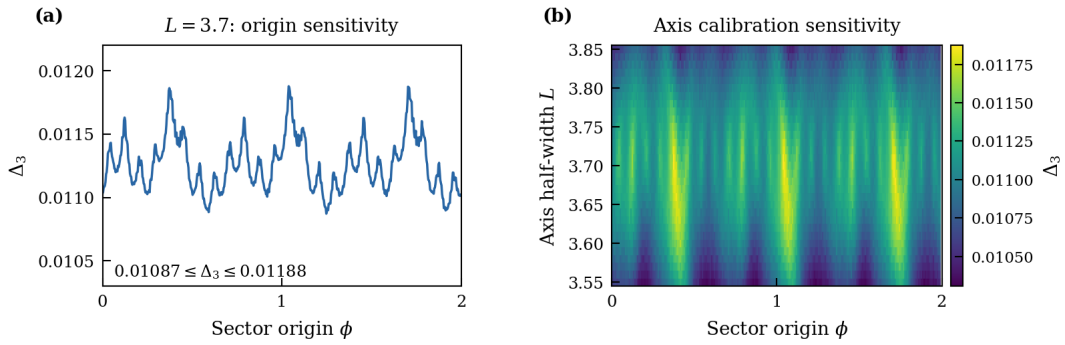


FIG. S2. Sensitivity of the histogram estimated from Ref. [17] to the momentum-sector origin and calibration. (a) The cubic quantity Δ_3 for 601 sector offsets $\phi \in [0, 2)$ at the axis half-width $L = 3.7$. (b) A scan of half-widths $L \in [3.55, 3.85]$, with offsets at each value. All sampled histograms satisfy the pairwise conditions and have $\Delta_3 > 0$.

S6. PAIR TRANSFER BEYOND FIXED ONE-BODY EVOLUTION

The previous section showed that an estimated experimental counting distribution can be compatible with fixed one-body scattering while requiring noncommuting terminal effects. We now consider the opposite situation. Can experimentally observed counting probabilities rule out fixed one-body evolution altogether? To address this question, we examine the pair-transfer experiment reported in Ref. [18].

The experiment confines opposite-spin ^{6}Li atoms in an optical-superlattice double well. Interaction pulses and a charge-sensitive tilt move both atoms together between the wells while suppressing changes in states with one atom in each well. We use the output probabilities in the lower panel of their Fig. 5(e), whose input is a left doublon with both atoms in the left well [18].

The four resolved one-particle modes are labeled by the well L, R and the spin $\uparrow, \downarrow$. The initial doublon occupies two orthogonal spin orbitals and is a Slater determinant:

$$|\Psi_{\text{in}}\rangle = |\uparrow\downarrow, 0\rangle = \hat{c}^\dagger_{L\uparrow}\hat{c}^\dagger_{L\downarrow}|0\rangle \ . \tag{S47}$$

Here, $\hat{c}^\dagger_{L\sigma}$ creates a particle of spin σ in the left well, and $|0\rangle$ is the vacuum.

Since the initial doublon is a Slater determinant, it satisfies the input-state assumption of our one-body criterion. To apply the three-terminal test, we distinguish the two spin modes in the left well while combining both spin modes in the right well into a single terminal. This choice keeps the relevant spin-resolved counting information and defines three terminals:

$$T_1 = \text{span}\{|L\uparrow\rangle\}, \quad T_2 = \text{span}\{|L\downarrow\rangle\}, \quad T_3 = \text{span}\{|R\uparrow\rangle, |R\downarrow\rangle\}. \tag{S48}$$

The three projectors are orthogonal and complete on these four modes. Their dimensions $(1, 1, 2)$ enforce $p_1 = p_2 = 0$ by Pauli exclusion. Table S1 gives the counting outcomes for all six two-particle configurations.

TABLE S1. Mapping from the resolved double-well configurations to the terminal counts of Eq. (S48). The comma in each ket separates the left and right wells; configurations in the same row contribute to the same probability.

Resolved configurations	Terminal counts (n_1, n_2, n_3)	Probability
$\|\uparrow\downarrow, 0\rangle$	$(1, 1, 0)$	s_{12}
$\|0, \uparrow\downarrow\rangle$	$(0, 0, 2)$	p_3
$\|\uparrow, \downarrow\rangle$, $\|\uparrow, \uparrow\rangle$	$(1, 0, 1)$	s_{13}
$\|\downarrow, \uparrow\rangle$, $\|\downarrow, \downarrow\rangle$	$(0, 1, 1)$	s_{23}

Before applying the criterion, we must ensure that the experimental probabilities form a complete counting distribution for the chosen terminals. In particular, renormalizing an incomplete subset of outcomes would change the counting problem and could lead to an incorrect realizability test.

We first check whether the four published opposite-spin frequencies account for all outcomes. Here, x is the numerical duration of the charge-sensitive tilt pulse in microseconds, as shown on the horizontal axis of the published

Fig. 5(e). At $x = 30, 50$, the four listed frequencies sum to unity at their reported precision; at every other published setting for this input, their sum is less than one. We select these two complete histograms on that basis, without renormalization. Table S2 gives the probabilities and their published individual 68% confidence intervals.

TABLE S2. Probabilities for the two complete pair-transfer histograms from Ref. [18]. The notation $f^{+u}_{-\ell}$ gives the reported frequency f and the lower and upper lengths ℓ, u of its individual 68% confidence interval. The two additional probabilities $p_1 = p_2 = 0$ follow from Pauli exclusion, since each of terminals 1 and 2 contains a single fermionic mode.

Probability	Pulse duration x (μs)	
	30	50
p_3	$0.3889^{+0.0583}_{-0.0553}$	$0.4098^{+0.0636}_{-0.0607}$
s_{12}	$0.5694^{+0.0567}_{-0.0586}$	$0.5574^{+0.0618}_{-0.0637}$
s_{13}	$0.0000^{+0.0135}_{-0.0000}$	$0.0328^{+0.0311}_{-0.0162}$
s_{23}	$0.0417^{+0.0303}_{-0.0179}$	$0.0000^{+0.0160}_{-0.0000}$

The chosen terminal partition considerably simplifies the realizability test. Because each left-well terminal contains only one fermionic mode, Pauli exclusion requires $p_1 = p_2 = 0$. Substituting these constraints into the general three-terminal criterion gives:

$$\Delta_{12} = s_{12}^2, \quad \Delta_{13} = s_{13}^2, \quad \Delta_{23} = s_{23}^2, \quad \Delta_3 = s_{12}(s_{13}s_{23} - p_3 s_{12})\,. \tag{S49}$$

All three conditional pair tests are satisfied automatically. For $s_{12} > 0$, a common one-body realization exists precisely when $s_{13}s_{23} \geq p_3 s_{12}$; at $s_{12} = 0$, this condition and the cubic inequality are automatic. An admissible histogram with $p_1 = p_2 = 0$ can be realized with effects of ranks at most $(1, 1, 2)$, so the criterion also respects the dimensions of the chosen experimental terminals. Now, the realizability question reduces to whether the probabilities of the two separated-particle outcomes are sufficiently large compared with those of the two intact-pair outcomes.

Evaluating Δ_3 with the published frequencies before rounding for Table S2 gives:

$$\Delta_{3,30} \simeq -0.1261, \quad \Delta_{3,50} \simeq -0.1273\,. \tag{S50}$$

For both pulse durations, the negative values of Δ_3 violate the only nontrivial one-body condition. Thus, the reported central histograms cannot be reproduced by a fixed one-body scatterer acting on the specified Slater input.

In both cases, appreciable probabilities are observed for the intact pair to occupy either well, whereas pair-separated outcomes are strongly suppressed. This imbalance is incompatible with fixed one-body evolution. In each, one separation frequency is zero, while both doublon frequencies are nonzero; consequently $s_{13}s_{23} < p_3 s_{12}$.

The violation of Δ_3 establishes the incompatibility with fixed one-body evolution, but it is useful to express this result as a more direct physical constraint. Specifically, if one-body evolution allows the intact pair to occupy either well, how much probability must it assign to configurations in which the particles are separated?

To answer this, we denote the left- and right-doublon probabilities by $D_L = s_{12}$ and $D_R = p_3$, and the two separation probabilities by $X = s_{13}$ and $Y = s_{23}$. The product condition, together with the arithmetic–geometric mean inequality, gives:

$$XY \geq D_L D_R, \quad P_{\text{sep}} := X + Y \geq 2\sqrt{D_L D_R}\,. \tag{S51}$$

Thus one-body evolution that assigns appreciable probability to both intact-pair outputs must also produce separated particles. Table S3 compares this minimum with the two reported histograms.

TABLE S3. Total separation probability and its one-body lower bound for the two pair-transfer histograms.

Pulse duration x (μs)	P_{sep}	$2\sqrt{D_L D_R}$
30	0.0417	0.9412
50	0.0328	0.9559

So far, the exclusion has been obtained using a three-terminal partition that resolves the two spin states in the left well. We now ask whether this spin resolution is essential, or whether the incompatibility with one-body evolution can already be detected by counting only the total number of particles in each well.

To test this, we merge the two left-well terminals and consider the resulting right-well occupation distribution. For the resulting right-well occupation distribution, a sum of two independent arrivals is possible precisely when the discriminant is nonnegative:

$$D_3 = (s_{13} + s_{23})^2 - 4p_3 s_{12} = P_{\mathrm{sep}}^2 - 4D_L D_R \,. \tag{S52}$$

The separation probabilities in Table S3 give $D_3 < 0$ for both datasets, so they also fail the test using only the total count in each well. The full three-terminal product condition in Eq. (S51) is stronger when $X \neq Y$.

The separation bound gives a necessary condition for one-body evolution. To show that this bound is tight, we construct an explicit one-body model that saturates it. Consider a spin-preserving splitter with the same transmission probability τ for both spins. Its occupied output orbitals are:

$$|\psi_\sigma\rangle = \sqrt{1-\tau}\,|L\sigma\rangle + \sqrt{\tau}\,|R\sigma\rangle \;, \quad \sigma \in \{\uparrow,\downarrow\}\,, \quad 0 \le \tau \le 1\,. \tag{S53}$$

These orthogonal orbitals form a Slater determinant with probabilities:

$$D_L = (1-\tau)^2, \quad D_R = \tau^2, \quad X = Y = \tau(1-\tau)\,, \quad P_{\mathrm{sep}} = 2\sqrt{D_L D_R}\,. \tag{S54}$$

It saturates both inequalities in Eq. (S51).

The one-body splitter illustrates how transferring an intact pair between two wells necessarily produces separated-particle outcomes. We now contrast this with ideal coherent pair transfer [18], in which the pair occupies a superposition of the two doublon configurations without populating separated states. For mixing angle θ and relative phase φ, its state is:

$$|\Psi(\theta)\rangle = \cos\theta\,|\uparrow\downarrow,0\rangle + \mathrm{e}^{\mathrm{i}\varphi}\sin\theta\,|0,\uparrow\downarrow\rangle\,, \quad u = \sin^2\theta\,. \tag{S55}$$

Here u is the probability that the intact pair has reached the right well. The counting distribution and its cubic criterion value are:

$$(p_1,p_2,p_3;s_{12},s_{13},s_{23}) = (0,0,u;1-u,0,0), \quad \Delta_3 = -u(1-u)^2\,. \tag{S56}$$

Every $0 < u < 1$ is excluded by the fixed one-body criterion, with a minimum $\Delta_3 = -4/27 \simeq -0.1481$ at $u = 1/3$. The deterministic endpoints are realizable. Because Eq. (S56) is independent of φ, the occupation measurement does not determine the phase coherence between the two doublon configurations. Figure S3 summarizes the terminal partitions, the measured or estimated probabilities, and the separation-bound comparison.

These results show that fixed one-body evolution imposes a minimum amount of pair separation whenever both intact-pair outcomes have nonzero probabilities. The reported pair-transfer histograms violate this requirement, and the ideal pair-transfer model illustrates the same incompatibility for any nontrivial superposition of the two doublon configurations. In these cases, the exclusion is already visible without spin resolution.

S7. WHY SEPARATE TERMINAL TESTS DO NOT ESTABLISH A COMMON SCATTERER

The charge-counting experiment of Ref. [19] counts how often a quantum dot discharges during a measurement interval. From the count probabilities, the authors calculate quantities whose signs are constrained for independent single-particle transfers; a violation excludes this description. For our fixed-N input, the same test applies to the occupation of each terminal separately. We ask whether a common scatterer follows when every terminal has a complete independent-event description, which in particular passes these tests at every order.

To state these separate tests, let $g_i(z)$ be the generating polynomial for the count n_i in terminal i alone, with z an auxiliary counting variable. For N independent single-particle events it has the form [13, 14]:

$$g_i(z) = \prod_{r=1}^{N} (1 - \nu_{ir} + \nu_{ir} z), \quad 0 \le \nu_{ir} \le 1\,. \tag{S57}$$

Each factor describes independent event r, where it contributes one count with probability ν_{ir} and zero otherwise.

In a one-body realization, the ν_{ir} are the eigenvalues of A_i. To derive the sign test, we differentiate the logarithm of $g_i(z)$; the derivative of integer order $q \ge 1$ is the factorial cumulant [20, 21]:

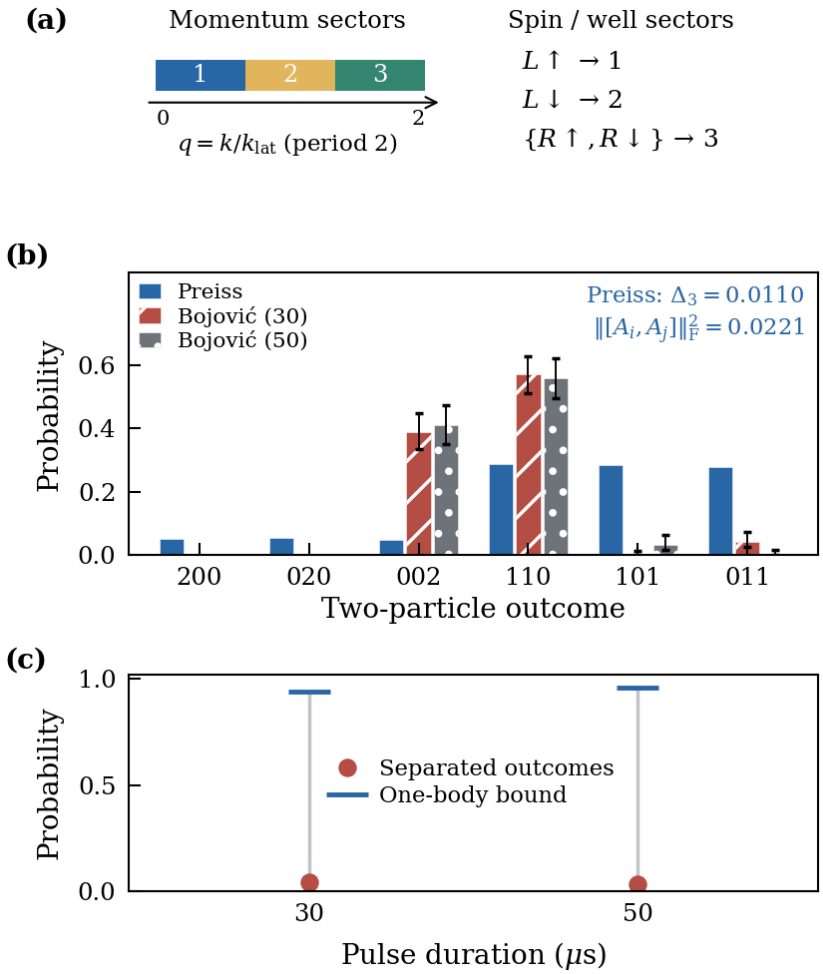


FIG. S3. Terminal partitions and counting constraints for the two experimental examples. (a) The periodic momentum sectors for Ref. [17], shown over one period of $q = k/k_{\mathrm{lat}}$, and the spin/well partition for Ref. [18]. (b) The six-outcome histograms also shown in main-text Fig. 2(a). Blue denotes the estimate from the published Preiss correlation map; red and gray denote the published Bojović distributions at $x = 30, 50$. The annotated commutator norm is inferred from the estimated Preiss histogram through Eq. (S24). Black error bars are the published individual 68% confidence intervals for the Bojović probabilities; the 200 and 020 outcomes are forbidden by Pauli exclusion. (c) The separation probabilities $P_{\mathrm{sep}} = s_{13} + s_{23}$ and the corresponding one-body lower bounds $2\sqrt{p_3 s_{12}}$ computed from the reported frequencies.

$$C_{\mathrm{F},q,i} = \left.\frac{\mathrm{d}^q \ln g_i(z)}{\mathrm{d}z^q}\right|_{z=1} = (-1)^{q-1}(q-1)! \sum_{r=1}^{N} \nu_{ir}^q \,. \tag{S58}$$

Thus $(-1)^{q-1} C_{\mathrm{F},q,i} \geq 0$ is necessary for independent events. The complementary count $N - n_i$ has probabilities $1 - \nu_{ir}$ and obeys the same sign rule. When $C_{\mathrm{F},1,i} > 0$, the bound $\nu_{ir}^q \leq \nu_{ir}$ also gives $(-1)^{q-1} C_{\mathrm{F},q,i}/[(q-1)! C_{\mathrm{F},1,i}] \leq 1$ [19].

The factorization determines the statistics of each count separately. A common scatterer must also supply terminal matrices that sum to the identity on the same occupied orbital space. We test this additional requirement at equal mean occupations using the point X in Fig. S1(c). Its six probabilities are:

$$(p_1, p_2, p_3; s_{12}, s_{13}, s_{23}) = \frac{1}{36}(0, 4, 1; 9, 15, 7) \,. \tag{S59}$$

Every terminal has mean occupation $\mu_i = 2/3$. For terminals 1, 2, and 3, the independent-event probabilities are $(0, 2/3)$, $(1/3, 1/3)$, and $((2+\sqrt{3})/6, (2-\sqrt{3})/6)$, respectively. Each terminal and its complementary count therefore obey the factorial-cumulant sign rule at every order.

To check whether the separate descriptions fit together, we evaluate the four joint conditions:

$$(\Delta_{12}, \Delta_{13}, \Delta_{23}; \Delta_3) = \left(\frac{1}{16}, \frac{25}{144}, \frac{11}{432}; -\frac{1}{1296}\right) . \tag{S60}$$

All three conditional pair tests pass, but $\Delta_3 < 0$ excludes a common scatterer. The cross-covariance sign test also admits this distribution:

$$(\mathrm{Cov}(n_1, n_2), \mathrm{Cov}(n_1, n_3), \mathrm{Cov}(n_2, n_3)) = \left(-\frac{7}{36}, -\frac{1}{36}, -\frac{1}{4}\right) . \tag{S61}$$

To see why no one-body scatterer can produce these counts, suppose that a common realization exists. Terminal 2 would have $a_2 = 1/3$ and $p_2 = 1/9$, forcing $A_2 = \mathbb{1}_2/3$. Terminal 1 has $a_1 = 1/3$ and $p_1 = 0$, so the eigenvalues of A_1 would be $2/3$ and zero. The condition $A_1 + A_2 + A_3 = \mathbb{1}_2$ would then fix:

$$A_3 = \frac{2}{3}\mathbb{1}_2 - A_1, \quad \mathrm{spec}\, A_3 = \{0, 2/3\}, \quad \det A_3 = 0 . \tag{S62}$$

Here spec A_3 denotes the eigenvalues of A_3. This contradicts the required probability $p_3 = 1/36$.

The preceding example has a zero counting probability and a repeated single-terminal eigenvalue. To check whether incompatibility persists without either feature, consider the normalized family with a real parameter $0 < x < 1/90$:

$$\begin{aligned} (p_1, p_2, p_3) &= (1/9 - 10x, 1/9 - 4x, 1/9 - x) , \\ (s_{12}, s_{13}, s_{23}) &= (2/9 + 13x, 2/9 + 7x, 2/9 - 5x) \end{aligned} . \tag{S63}$$

All six probabilities in Eq. (S63) are positive, and every terminal again has mean occupation $2/3$. The two independent-event probabilities for terminal i are:

$$\nu_{i,\pm} = \frac{1}{3} \pm \sqrt{k_i x}, \quad (k_1, k_2, k_3) = (10, 4, 1) . \tag{S64}$$

The constants k_i specify the three terminal distributions. All six $\nu_{i,\pm}$ lie strictly between zero and one. By Eq. (S58), each terminal and its complementary count obey the sign rule and the normalized bound at every order. Testing the three terminal descriptions together gives:

$$(\Delta_{12}, \Delta_{13}, \Delta_{23}) = (12x + 9x^2, 8x + 9x^2, 9x^2) > 0 , \tag{S65}$$

$$\Delta_3 = -9x^2 < 0 . \tag{S66}$$

A common realization would require Bloch-vector lengths $\sqrt{x}(\sqrt{10}, 2, 1)$. They cannot form a triangle because $\sqrt{10} > 2 + 1$. The three separate independent-event descriptions cannot share one set of occupied orbitals. All probabilities, the single-terminal discriminants D_i, and the three Δ_{ij} are strictly positive, while $\Delta_3 < 0$, so these conditions persist under small changes of the normalized probabilities.

For two particles, the complete single-terminal distributions determine all six joint probabilities: they give $p_i = P(n_i = 2)$ and $b_i = P(n_i = 1)$, from which $s_{ij} = (b_i + b_j - b_k)/2$ for distinct i, j, k. Separate tests do not require the three terminal matrices to act on the same occupied space and sum to the identity. The four inequalities test whether all terminals admit the same one-body realization and guarantee its existence when they hold.

S8. TESTS BASED ON A RANDOMLY SELECTED PARTICLE PAIR

For more than two particles, mean occupations and pair correlations do not determine the full counting histogram. We derive necessary one-body constraints from these measurements by selecting a pair uniformly from each recorded event. The resulting two-particle distribution can be calculated from the count records without physically removing particles.

Consider m terminals and a fixed number $N \geq 2$ of particles. For an event with terminal occupations $\boldsymbol{n} = (n_1, \ldots, n_m)$, each of its $\binom{N}{2}$ unordered particle pairs has the same chance of being selected. Let $\boldsymbol{k} = (k_1, \ldots, k_m)$ record the terminal occupations of the retained pair. There are $\prod_i \binom{n_i}{k_i}$ ways to choose that pair from the event, so the conditional probability is:

$$P(\boldsymbol{k} \mid \boldsymbol{n}) = \frac{\prod_i \binom{n_i}{k_i}}{\binom{N}{2}}, \quad \sum_i k_i = 2 . \tag{S67}$$

Averaging over events gives the probabilities $\tilde{p}_i$ for both selected particles to reach terminal i and $\tilde{s}_{ij}$ for one to reach each of distinct terminals i, j. They depend only on the measured second factorial moments:

$$F_i = \langle n_i(n_i-1)\rangle\,, \quad H_{ij} = \langle n_i n_j\rangle\,, \quad \tilde{p}_i = \frac{F_i}{N(N-1)}, \quad \tilde{s}_{ij} = \frac{2H_{ij}}{N(N-1)}\,. \tag{S68}$$

These probabilities sum to one. Their mean occupations are $\tilde{\mu}_i = \langle k_i\rangle = 2\mu_i/N$, where $\mu_i = \langle n_i\rangle$ is the original N-particle mean. Thus the measured pair weights give both the probabilities and the means of the selected-pair distribution.

To check whether pair selection preserves real stability, we express it through derivatives of the original polynomial χ_N. Let $D = \sum_i \partial_i$, with $\partial_i = \partial/\partial z_i$, differentiate with respect to a common shift of all counting variables. The generating polynomial $\tilde{\chi}_2$ of the selected pair is:

$$\tilde{\chi}_2(\boldsymbol{z}) = \frac{2!}{N!} D^{N-2}\chi_N(\boldsymbol{z})\,. \tag{S69}$$

For an individual event, use $\boldsymbol{z}^{\boldsymbol{n}} = \prod_i z_i^{n_i}$ and $|\boldsymbol{k}| = \sum_i k_i$.
Expanding the repeated derivative gives exactly the pair-selection weights:

$$\frac{2!}{N!} D^{N-2}\boldsymbol{z}^{\boldsymbol{n}} = \sum_{|\boldsymbol{k}|=2} \frac{\prod_i \binom{n_i}{k_i}}{\binom{N}{2}} \boldsymbol{z}^{\boldsymbol{k}}\,. \tag{S70}$$

The derivative therefore gives the normalized pair-selection probabilities, proving Eq. (S69).

Pair selection preserves real stability for any real-stable χ_N. To prove this, introduce an extra complex variable t in $\chi_N(z_1+t,\ldots,z_m+t)$. This polynomial is stable because the imaginary part of every shifted argument is positive when $\operatorname{Im} z_i > 0$ and $\operatorname{Im} t > 0$. Differentiate $N-2$ times with respect to t and then set $t=0$. Both operations preserve stability or give the zero polynomial [7, 10, 22]. The selected-pair polynomial is normalized, so it cannot be zero. Thus the selected-pair distribution is real stable. For three terminals, the complete realizability theorem then supplies a two-particle one-body realization of the selected-pair distribution.

The four two-particle inequalities therefore give necessary constraints on the measured moments of the original distribution. For $m=3$, substitution of Eq. (S68) into Eq. (S20) and cancellation of their common positive factors gives:

$$H_{ij}^2 \geq F_i F_j \quad (1 \leq i < j \leq 3)\,, \tag{S71}$$

$$F_1F_2F_3 + 2H_{12}H_{13}H_{23} \geq F_1H_{23}^2 + F_2H_{13}^2 + F_3H_{12}^2\,. \tag{S72}$$

The single-terminal probability bound $\tilde{p}_i \leq (\tilde{\mu}_i/2)^2$ similarly gives:

$$F_i \leq \frac{N-1}{N}\mu_i^2\,. \tag{S73}$$

Failure of any of these inequalities excludes every fixed one-body realization of the full N-particle histogram.

Passing these pair tests does not guarantee a one-body realization of the full histogram.
An example is the following normalized three-particle polynomial, together with its selected-pair polynomial obtained by applying $D/3$:

$$\chi_3(\boldsymbol{z}) = \frac{5}{6}z_1z_2z_3 + \frac{z_1^3+z_2^3+z_3^3}{18}, \quad \tilde{\chi}_2(\boldsymbol{z}) = \frac{z_1^2+z_2^2+z_3^2}{18} + \frac{5}{18}(z_1z_2+z_1z_3+z_2z_3)\,. \tag{S74}$$

For the selected-pair distribution, every $\tilde{\Delta}_{ij} = 7/108 > 0$ and $\Delta_3 = 1/108 > 0$, so it is realizable. The original cubic must also have only real roots along every line $\boldsymbol{x} + t\boldsymbol{1}$ with real $\boldsymbol{x}$ if it is to be real stable. The choice $\boldsymbol{x} = (1,1,-2)^T$ gives:

$$\chi_3(1+t,1+t,-2+t) = t^3 - \frac{3}{2}t - 2\,. \tag{S75}$$

Its negative discriminant $-189/2$ implies two nonreal roots. The root in the upper half-plane violates real stability of χ_3. The full three-particle distribution is incompatible with one-body dynamics although its selected-pair distribution is compatible.

The selected-pair model uses two particles. To show that the original first and second moments can all have a one-body explanation at the original particle number $N=3$, we construct a three-particle model that matches them

exactly. Write the second factorial moments as a matrix with entries $M_{ij} = \langle n_i(n_j - \delta_{ij})\rangle$. Its diagonal entries count ordered pairs in the same terminal, and its off-diagonal entries equal $\langle n_i n_j\rangle$.

For this cubic the first and second factorial moments are:

$$\boldsymbol{\mu}_3 = \begin{bmatrix}1\\1\\1\end{bmatrix}, \quad M_3 = \begin{bmatrix}1/3 & 5/6 & 5/6\\ 5/6 & 1/3 & 5/6\\ 5/6 & 5/6 & 1/3\end{bmatrix}. \tag{S76}$$

Use three independent single-particle arrivals to reproduce these moments. Let $(\Pi_3)_{ir}$ be the probability for channel r to reach terminal i, so each column must sum to one. A suitable choice is:

$$v = \frac{1+\sqrt{2}}{3}, \quad u = \frac{2-\sqrt{2}}{6}, \quad \Pi_3 = \begin{bmatrix}v & u & u\\ u & v & u\\ u & u & v\end{bmatrix}, \quad v + 2u = 1\,. \tag{S77}$$

For independent arrivals, the means are the row sums of Π_3, while the second factorial moments are $\boldsymbol{\mu}_3\boldsymbol{\mu}_3^T - \Pi_3\Pi_3^T$. The chosen entries are positive, and direct multiplication confirms that the arrivals are normalized and have the required moments:

$$\Pi_3^T\mathbf{1} = \mathbf{1}, \quad \Pi_3\mathbf{1} = \boldsymbol{\mu}_3, \quad \Pi_3\Pi_3^T = \boldsymbol{\mu}_3\boldsymbol{\mu}_3^T - M_3\,. \tag{S78}$$

To turn these arrival probabilities into a one-body model, take the diagonal effects $(A_{3,*})_i = \mathrm{diag}((\Pi_3)_{i1}, (\Pi_3)_{i2}, (\Pi_3)_{i3})$. They are positive and sum to $\mathbb{1}_3$, so Eq. (S3) supplies a fixed scatterer with generating polynomial:

$$\chi_{3,*}(\boldsymbol{z}) = \prod_{r=1}^{3}\left(vz_r + u\sum_{i\neq r} z_i\right). \tag{S79}$$

These identities give $\chi_{3,*}$ the same first and second factorial moments as χ_3, and therefore the same ordinary second moments and selected-pair distribution. Events in which all three particles reach one terminal distinguish the full histograms. Writing $P_\chi(300)$ for the probability of $\boldsymbol{n} = (3,0,0)$ under a polynomial χ, the two predictions are:

$$P_{\chi_3}(300) = \frac{1}{18}, \quad P_{\chi_{3,*}}(300) = vu^2 = \frac{\sqrt{2}-1}{54} \simeq 0.00767062\,. \tag{S80}$$

The events 030 and 003 have the same respective probabilities by symmetry. Figure S4 compares the complete event groups. Because a one-body model reproduces these moments exactly, no test of first and second moments alone can reject χ_3. The nonreal roots of the full counting polynomial do reject it.

S9. THE FOUR-TERMINAL VÁMOS DISTRIBUTION

With four terminals, a real-stable counting distribution can have no one-body realization [23, 24]. We prove this for a four-particle distribution whose separate three-terminal mergings and selected-pair distribution each admit a one-body model. We also construct a four-particle one-body model that matches all its first and second moments, so those measurements alone cannot exclude one-body evolution.

For four counting variables $\boldsymbol{z} = (z_1, z_2, z_3, z_4)$, define χ_V by the specialized Vámos polynomial h_4 [24]:

$$h_4(\boldsymbol{z}) = z_1^2z_2^2 + 4(z_1+z_2+z_3+z_4)(z_1z_2z_3 + z_1z_2z_4 + z_1z_3z_4 + z_2z_3z_4), \quad \chi_V = \frac{h_4}{65}\,. \tag{S81}$$

It assigns probability $1/65$ to $(2,2,0,0)$, probability $4/65$ to each of the twelve distinct permutations of $(2,1,1,0)$, and probability $16/65$ to $(1,1,1,1)$. Every other four-particle bin has probability zero. All coefficients are nonnegative and their sum is $(1 + 12\times 4 + 16)/65 = 1$.

To establish that χ_V passes the real-stability condition, start from the known hyperbolicity of h_4 with respect to $\boldsymbol{e} = (1,1,0,0)^T$ [24]. By the definition in Sec. S2, $h_4(\boldsymbol{x} + t\boldsymbol{e})$ has only real roots in t for every real vector $\boldsymbol{x}$. Real stability requires this property for every direction $\boldsymbol{y}$ whose four entries are strictly positive.

The real-root property extends from $\boldsymbol{e}$ to these positive directions because h_4 stays nonzero between them. Along the line segment from $\boldsymbol{e}$ to $\boldsymbol{y}$, the first two coordinates remain positive and the other two remain nonnegative. The monomial $z_1^2z_2^2$ is strictly positive there and all remaining terms are nonnegative. Thus the segment remains in the connected component of $\{\boldsymbol{x} : h_4(\boldsymbol{x}) \neq 0\}$ containing $\boldsymbol{e}$. This component is the open hyperbolicity cone, and every

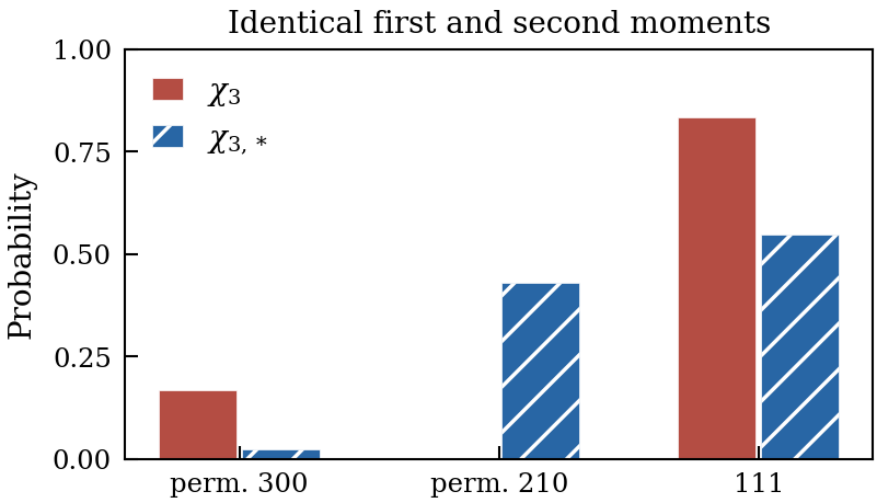


FIG. S4. Three-particle counting beyond first and second moments. The nonrealizable cubic χ_3 and the one-body model $\chi_{3,*}$ of Eq. (S79) have identical first and second counting moments. The three disjoint event groups contain the three permutations of 300, the six permutations of 210, and 111, respectively. Their probabilities are $(1/6, 0, 5/6)$ for χ_3 and $((\sqrt{2}-1)/18, (4-\sqrt{2})/6, (7+2\sqrt{2})/18)$ for $\chi_{3,*}$.

vector in it is a hyperbolicity direction [7, 10]. If $h_4(\boldsymbol{x}+\mathrm{i}\boldsymbol{y})$ vanished for real $\boldsymbol{x}$ and positive $\boldsymbol{y}$, then $t=\mathrm{i}$ would be a nonreal root of $h_4(\boldsymbol{x}+t\boldsymbol{y})$, a contradiction. Both h_4 and χ_V are real stable.

To rule out a common one-body realization, we use the largest occupation that occurs with nonzero probability in each subset of terminals. For a common one-body model to exist, all these largest counts must be dimensions of subspaces within the same occupied-orbital space [23]. Suppose, for a contradiction, that 4×4 Hermitian effects $A_i \succeq 0$ with $\sum_i A_i = \mathbb{1}_4$ realize χ_V. For a nonempty subset $J\subseteq\{1,2,3,4\}$, let r_J be the largest count with nonzero probability in that subset and A_J its combined effect:

$$r_J = \max_{\boldsymbol{n}:P(\boldsymbol{n})>0} \sum_{i\in J} n_i, \quad A_J = \sum_{i\in J} A_i\,. \tag{S82}$$

The largest count is the degree of the polynomial that counts particles in J. To obtain this polynomial, retain a counting variable t for terminals in J and set the others to one. For any fixed particle number N, the one-body determinant then gives:

$$\chi_N(t \text{ on } J; 1 \text{ elsewhere}) = \det[\mathbb{1}_N + (t-1)A_J]\,. \tag{S83}$$

Because $0\preceq A_J \preceq \mathbb{1}_N$, each nonzero eigenvalue contributes one linear factor in t, and every zero eigenvalue contributes the constant one. The degree is therefore the number of positive eigenvalues, or $\operatorname{rank} A_J$. Nonnegative counting probabilities identify the same degree with r_J.

The rank also gives a subspace dimension. Let $U_i = \operatorname{ran} A_i$ be the range of A_i in the occupied-orbital space. Positivity gives $\ker A_J = \bigcap_{i\in J} \ker A_i$, where ker denotes the null space. Taking orthogonal complements identifies the range of A_J with the span of the U_i for $i\in J$, denoted by $\sum_{i\in J} U_i$. Consequently:

$$r_J = \operatorname{rank} A_J = \dim\left(\sum_{i\in J} U_i\right), \quad U_i = \operatorname{ran} A_i \subseteq \mathbb{C}^N\,. \tag{S84}$$

We can therefore test whether one family of subspaces can have all the dimensions required by the nonzero counting probabilities [23]. For χ_V, these subspaces would all lie in $\mathbb{C}^4$. Using r_{ij} and r_{ijk} as shorthand for subsets $\{i,j\}$ and $\{i,j,k\}$, the nonzero-probability events in Eq. (S81) require:

$$r_i = 2, \quad r_{12} = 4, \quad r_{13} = r_{14} = r_{23} = r_{24} = r_{34} = 3, \quad r_{123} = r_{124} = r_{134} = r_{234} = 4\,. \tag{S85}$$

Subspaces of one common vector space must satisfy the Ingleton dimension inequality [23]:

$$r_{34} + r_{13} + r_{14} + r_{23} + r_{24} \geq r_3 + r_4 + r_{12} + r_{134} + r_{234}\,. \tag{S86}$$

The required dimensions make the left side 15 and the right side 16, contradicting the inequality.

A direct argument shows how the required subspace overlaps conflict, without invoking the general dimension inequality. The two-dimensional spaces U_1 and U_2 are complementary because $r_{12} = 4$. The ranks $r_{13} = r_{23} = 3$ imply that U_3 intersects each of them in one-dimensional subspaces $L_{31} = U_3 \cap U_1$ and $L_{32} = U_3 \cap U_2$. Because $U_1 \cap U_2 = \{0\}$, these lines span $U_3 = L_{31} \oplus L_{32}$. Likewise, with $L_{4i} = U_4 \cap U_i$ for $i = 1, 2$, one has $U_4 = L_{41} \oplus L_{42}$. The requirement $r_{34} = 3$ then reads:

$$\dim(L_{31} + L_{41}) + \dim(L_{32} + L_{42}) = 3\,. \tag{S87}$$

Each summand is one or two, so exactly one pair of lines coincides. If $L_{31} = L_{41}$, then $r_{234} = 3$; if $L_{32} = L_{42}$, then $r_{134} = 3$. Both conclusions contradict the required dimensions for triples of terminals. The proof holds for complex subspaces and therefore excludes Hermitian terminal effects directly. It uses exactly which events have zero probability in the theoretical distribution; zero occurrences in a finite sample do not establish a zero probability.

A separate three-terminal test combines at least two of the four terminal labels. It can reject χ_V only if the resulting three-terminal polynomial fails real stability. Let a map $\pi : \{1, 2, 3, 4\} \to \{1, 2, 3\}$ assign each original terminal to one of three combined terminals, whose counting variables are w_1, w_2, w_3. The merged generating polynomial is:

$$\chi_{V,\pi}(w_1, w_2, w_3) = \chi_V(w_{\pi(1)}, w_{\pi(2)}, w_{\pi(3)}, w_{\pi(4)})\,. \tag{S88}$$

Setting counting variables equal preserves nonnegative coefficients, normalization, and homogeneity. It also preserves stability because each substituted variable still has positive imaginary part [10]. The three-terminal theorem then guarantees a four-particle one-body realization of every three-terminal merging.
For example, merging terminals 3 and 4 gives:

$$\chi_V(x, y, z, z) = \frac{x^2y^2 + 8x^2yz + 8xy^2z + 4x^2z^2 + 4y^2z^2 + 24xyz^2 + 8xz^3 + 8yz^3}{65}\,. \tag{S89}$$

The coefficients sum to 65/65. Each merging may have a different one-body realization. The six merged histograms together, however, retain enough information to recover the full four-terminal polynomial. In fact, they determine any four-variable polynomial of degree at most four uniquely. If two such polynomials f and g have identical mergings, their difference vanishes on every hyperplane $z_i = z_j$. Each linear factor $z_i - z_j$ therefore divides the difference, and their product has degree six:

$$\prod_{1 \leq i < j \leq 4} (z_i - z_j) \quad \text{divides} \quad f(\boldsymbol{z}) - g(\boldsymbol{z})\,. \tag{S90}$$

Since the difference has degree at most four, it must vanish. Consequently, the individual realizations of the six mergings of χ_V cannot all arise from one common four-terminal model.

Selecting a particle pair retains all four terminal labels. To check whether this pair information excludes χ_V, we test the complete selected-pair distribution. For $N = 4$, Eq. (S69) gives its polynomial $\tilde{\chi}_{V,2} = D^2\chi_V/12$. Direct differentiation yields:

$$\tilde{\chi}_{V,2} = \frac{26z_1^2 + 26z_2^2 + 24z_3^2 + 24z_4^2 + 120z_1z_2 + 112(z_1z_3 + z_1z_4 + z_2z_3 + z_2z_4 + z_3z_4)}{780}\,. \tag{S91}$$

The normalization is $(26 + 26 + 24 + 24 + 120 + 5 \times 112)/780 = 1$.
The two-particle criterion in Eq. (S17) uses the coefficient matrix G_V. Its row sums form $\boldsymbol{a}_V$, whose entries are half the mean counts of the selected pair:

$$G_V = \frac{1}{390}\begin{bmatrix} 13 & 30 & 28 & 28 \\ 30 & 13 & 28 & 28 \\ 28 & 28 & 12 & 28 \\ 28 & 28 & 28 & 12 \end{bmatrix}, \quad \boldsymbol{a}_V = G_V\mathbf{1} = \frac{1}{130}\begin{bmatrix} 33 \\ 33 \\ 32 \\ 32 \end{bmatrix}. \tag{S92}$$

A one-body realization exists exactly when $R_V \succeq 0$ and $\operatorname{rank} R_V \leq 3$, with:

$$R_V = \boldsymbol{a}_V\boldsymbol{a}_V^T - G_V = \frac{1}{50700}\begin{bmatrix} 1577 & -633 & -472 & -472 \\ -633 & 1577 & -472 & -472 \\ -472 & -472 & 1512 & -568 \\ -472 & -472 & -568 & 1512 \end{bmatrix}. \tag{S93}$$

The eigenvalues determine whether both conditions hold. The orthonormal vectors $(1,1,1,1)^T/2$, $(1,1,-1,-1)^T/2$, $(1,-1,0,0)^T/\sqrt{2}$, and $(0,0,1,-1)^T/\sqrt{2}$ are eigenvectors with respective eigenvalues:

$$0, \quad \frac{472}{12675}, \quad \frac{17}{390}, \quad \frac{8}{195}. \tag{S94}$$

The matrix satisfies $R_V \succeq 0$ and $\operatorname{rank} R_V = 3$. The general two-particle criterion in Eq. (S17) proves realizability of the selected-pair distribution with all four terminal labels retained. To construct the terminal effects, use the following Bloch vectors with positive real constants α, β, and γ:

$$\alpha = \sqrt{\frac{118}{12675}}, \quad \beta = \sqrt{\frac{17}{780}}, \quad \gamma = \sqrt{\frac{4}{195}}, \quad \boldsymbol{r}_1 = (\alpha,\beta,0), \quad \boldsymbol{r}_2 = (\alpha,-\beta,0), \quad \boldsymbol{r}_3 = (-\alpha,0,\gamma), \quad \boldsymbol{r}_4 = (-\alpha,0,-\gamma). \tag{S95}$$

These vectors have Gram matrix R_V, sum to zero, and satisfy $a_{V,i}^2 - |\boldsymbol{r}_i|^2 = (G_V)_{ii} \geq 0$. The resulting effects $A_i = a_{V,i}\mathbb{1}_2 + \boldsymbol{r}_i \cdot \boldsymbol{\sigma}$ realize Eq. (S91) by the orbital construction in Eq. (S3).

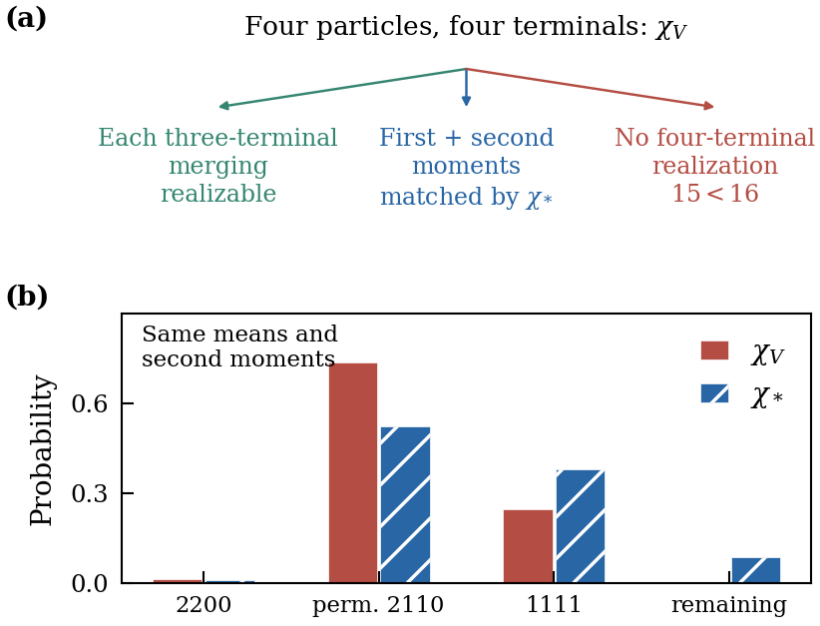


FIG. S5. A four-terminal distribution with no one-body realization and a model with matching moments. (a) Each three-terminal merging and the complete selected-pair distribution of χ_V admits a one-body realization; the four-particle model χ_* also reproduces all first and second counting moments. These separate realizations need not arise from a common four-terminal model. (b) Resolved events distinguish χ_V from the explicit model in Eq. (S101). The groups are 2200, all twelve permutations of 2110, 1111, and the remaining four-particle events. Each set of bars sums to one. No common set of occupied orbitals can have the subspace dimensions required by χ_V; this exclusion does not depend on the reference model.

The selected-pair model uses two particles. To check whether all the original first and second moments can have a one-body explanation at particle number $N = 4$, we construct a four-particle model that matches them exactly. This model uses four independent arrival channels.

The moments to be matched are the mean vector $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_4)^T$ and the factorial-moment matrix M, with entries:

$$\mu_i = \langle n_i \rangle, \quad M_{ij} = \langle n_i(n_j - \delta_{ij}) \rangle. \tag{S96}$$

Differentiating Eq. (S81) at $\boldsymbol{z} = \mathbf{1}$ gives:

$$\boldsymbol{\mu} = \frac{1}{65}\begin{bmatrix} 66 \\ 66 \\ 64 \\ 64 \end{bmatrix}, \quad M = \frac{1}{65}\begin{bmatrix} 26 & 60 & 56 & 56 \\ 60 & 26 & 56 & 56 \\ 56 & 56 & 24 & 56 \\ 56 & 56 & 56 & 24 \end{bmatrix}. \tag{S97}$$

For the four-particle model, use the following parameters, distinguished by a subscript star from those of the selected-pair model:

$$\alpha_* = \frac{16-\sqrt{118}}{65}, \quad \beta_* = \alpha_* + \frac{1}{130}, \quad \gamma_* = \sqrt{\frac{34}{65}}, \quad \eta_* = \sqrt{\frac{32}{65}}\,. \tag{S98}$$

Let Π_{ir} be the probability for independent channel r to reach terminal i, for $i, r = 1, 2, 3, 4$. The following choice will reproduce the target moments:

$$\Pi = \begin{bmatrix} \frac{1-2\alpha_*+\gamma_*}{2} & \frac{1-2\alpha_*-\gamma_*}{2} & \beta_* & \beta_* \\ \frac{1-2\alpha_*-\gamma_*}{2} & \frac{1-2\alpha_*+\gamma_*}{2} & \beta_* & \beta_* \\ \alpha_* & \alpha_* & \frac{1-2\beta_*+\eta_*}{2} & \frac{1-2\beta_*-\eta_*}{2} \\ \alpha_* & \alpha_* & \frac{1-2\beta_*-\eta_*}{2} & \frac{1-2\beta_*+\eta_*}{2} \end{bmatrix}. \tag{S99}$$

The bounds $10 < \sqrt{118} < 11$ and $0 < \gamma_*, \eta_* < 3/4$ ensure that every entry is strictly positive. Each column must sum to one because each channel carries one particle. The row sums give the mean occupations, and independence gives the second factorial moments as $\boldsymbol{\mu}\boldsymbol{\mu}^T - \Pi\Pi^T$. Direct substitution confirms normalization and the required moments:

$$\Pi^T \mathbf{1} = \mathbf{1}, \quad \Pi\mathbf{1} = \boldsymbol{\mu}, \quad \Pi\Pi^T = \boldsymbol{\mu}\boldsymbol{\mu}^T - M\,. \tag{S100}$$

To realize these channel probabilities with a fixed fermion scatterer, use the diagonal terminal effects A_i^*. Their determinant gives the full four-particle polynomial χ_*:

$$A_i^* = \mathrm{diag}(\Pi_{i1}, \Pi_{i2}, \Pi_{i3}, \Pi_{i4}), \quad \chi_*(\boldsymbol{z}) = \det\left(\sum_{i=1}^4 z_i A_i^*\right) = \prod_{r=1}^4 \left(\sum_{i=1}^4 \Pi_{ir} z_i\right)\,. \tag{S101}$$

These effects are positive semidefinite and sum to $\mathbb{1}_4$. With terminal mode dimensions unrestricted, the orbital construction in Eq. (S3) gives a fixed one-body scatterer for this four-particle model. The output orbitals can also be written explicitly. Choose mutually orthogonal modes $|i, r\rangle$, with four modes labeled $r = 1, 2, 3, 4$ in each terminal i. The occupied orbitals and output Slater state of this model can be chosen as:

$$|\psi_{*,r}\rangle = \sum_{i=1}^4 \sqrt{\Pi_{ir}}\,|i, r\rangle\,, \quad |\Psi_*\rangle = \prod_{r=1}^4 \hat{c}^\dagger(\psi_{*,r})\,|0\rangle\,. \tag{S102}$$

Column normalization of Π normalizes each orbital, while the mode labels r ensure mutual orthogonality. The resulting Slater state has first moments $\boldsymbol{\mu}_* = \Pi\mathbf{1} = \boldsymbol{\mu}$, and its second-order factorial moments satisfy:

$$(M_*)_{ij} = \mu_i\mu_j - \sum_{r=1}^4 \Pi_{ir}\Pi_{jr} = M_{ij}\,. \tag{S103}$$

Since $\langle n_i n_j\rangle = M_{ij} + \delta_{ij}\mu_i$, all ordinary second moments also agree. Thus, no criterion depending only on first- and second-order counting moments can distinguish χ_V from this fixed four-particle one-body model.

Real stability is therefore insufficient once four terminal counts are resolved. The separately realizable mergings and pair distribution, and the model with matching first and second moments, do not provide a common realization of the full histogram. The failure persists for larger terminal numbers: append empty terminals to the counterexample. A realization would force every added effect to vanish, since its mean count equals its nonnegative trace and is zero, leaving the forbidden four-terminal realization. Thus three is the largest terminal number for the stability–realizability equivalence to hold at arbitrary fixed particle number.

Figure S5 compares the resolved event probabilities of χ_V and the explicit model χ_*. The agreement of their first and second moments follows from Eq. (S100), while their full histograms differ.

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